

AMULYA KAUSHIK

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LinkedIn

GitHub

Portfolio

Education

Jaypee Institute of Information Technology

Noida, India

Bachelor of Technology - Computer Science; CGPA: 7.69

Jul 2023 – May 2027

- **Core Coursework:** Data Structures & Algorithms, Object Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering, Web Services.

Experience

Imarticus Learning

May 2026 – July 2026

Data Analytics Intern

Delhi, India

- Queried and analyzed **119K+ hotel records** using advanced **SQL** to derive business insights.
- Processed and visualized **4.6K+ space missions** using **Python, Pandas, and Matplotlib** to identify launch trends.
- Developed **ScreenShield**, an ML-based image classifier using FFT features and **Random Forest**, achieving **86%+ accuracy**.

Imagine Pharma Solutions

Mar 2026 – Apr 2026

Frontend Developer (Freelance)

Remote

- Designed and deployed a responsive landing page for a **pharmaceutical startup** using **React, Vite, and Tailwind CSS**.
- Implemented **15+ responsive UI sections** with smooth animations and routing using **Framer Motion** and **React Router**.
- Engineered **20+ reusable React components**, enabling scalable frontend development and simplified content updates.

Projects

Cloud Distributed Storage System [\[View Project\]](#) | Python, Flask, PostgreSQL, AWS S3

Mar 2026 – Apr 2026

- Built a forensic storage platform using **AES-256-GCM** and **SHA-256**, achieving **100% tamper detection** across 3 replicas.
- Implemented an immutable **Ed25519**-signed blockchain ledger for chain-of-custody with **100% audit tamper resistance**.
- Architected versioned REST APIs with a **4-tier RBAC** model for secure multi-tenant access control.
- Optimized metadata retrieval using indexed **PostgreSQL (Supabase)** and **AWS S3**, reducing query time by **40%**.

AI Digital Twin [\[View Project\]](#) | Python, FastAPI, PostgreSQL, ChromaDB, Docker, LLMs

Jun 2026 – Present

- Built a context-aware AI system that ingests documentation and maps semantic memories using vector embeddings.
- Configured a **500-token sliding-window** chunking pipeline with 100-token overlap for vector indexing in **ChromaDB**.
- Programmed a persona engine combining up to **20,000 characters** of memory into structured JSON profiles via **Pydantic**.
- Designed scalable backend orchestration APIs using **FastAPI, PostgreSQL**, and containerized services via **Docker**.

Ocean Level Rise Detection System [\[View Project\]](#) | Python, LSTM, XGBoost, SARIMA

Oct 2025 – Nov 2025

- Built a forecasting pipeline for **9 coastal cities** using **78K+ NOAA records** to predict sea-level trends and trigger alerts.
- Engineered ETL pipelines with lag and rolling features from NOAA APIs, processing **8.7K+ hourly records** per station.
- Benchmarked **LSTM, XGBoost and SARIMA**; **LSTM** achieved **90.46% accuracy** and **91.9%** lower error than SARIMA.

Technical Skills

Languages: C, C++, Python, JavaScript, HTML, CSS

Frameworks & Libraries: React, Node.js, Express.js, Flask, FastAPI, LangChain, Scikit-learn, TensorFlow

Database: MySQL, PostgreSQL, MongoDB, Redis, SQLite

API & Networking: RESTful APIs, WebSockets, TCP/IP Fundamentals

Developer Tools: Git, GitHub, Docker, Linux, Vercel, Streamlit, Jupyter Notebooks, Power BI

Skills: Data Structures & Algorithms, AI Agents & Agentic Workflows, Machine Learning, Deep Learning, GenAI

Achievements

Winner – BITBOX 6.0 Hackathon

Apr 2026

- Built a decentralized blockchain security platform using **static analysis and fuzzing** to find contract vulnerabilities.
- Designed threat monitoring infrastructure through mempool analysis and on-chain risk metrics to minimize attack surfaces.

Positions of Responsibility

Eloquence Literary Society, JIIT

Noida, India

Core Team Member

Sep 2023 – May 2025

- Led cross-functional coordination and event execution, driving clear reporting and documentation loops.

Parola, JIIT

Noida, India

Core Member – Management Team

May 2025 – Present

- Coordinated multi-tier logistics for **JOUST Fest** across distributed teams, managing workflows and partner operations.